Functions - Part 3: Exponential and Logarithmic Functions

Course: Pre-Calculus-I

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26CS(Evening) - Fall 2026

Outline

1 Exponential Functions

What is an Exponential Function?

Definition

An **exponential function** has the form:

$$f(x) = a^x$$

where $a > 0$ and $a \neq 1$. The constant a is called the **base**.

Key Characteristics:

- The variable x is in the **exponent**.
- Base a is a positive constant.
- Domain: $\mathbb{R} = (-\infty, \infty)$.
- Range: $(0, \infty)$ — always positive.
- y-intercept: $(0, 1)$ since $a^0 = 1$.
- Horizontal asymptote: $y = 0$ (the x-axis).

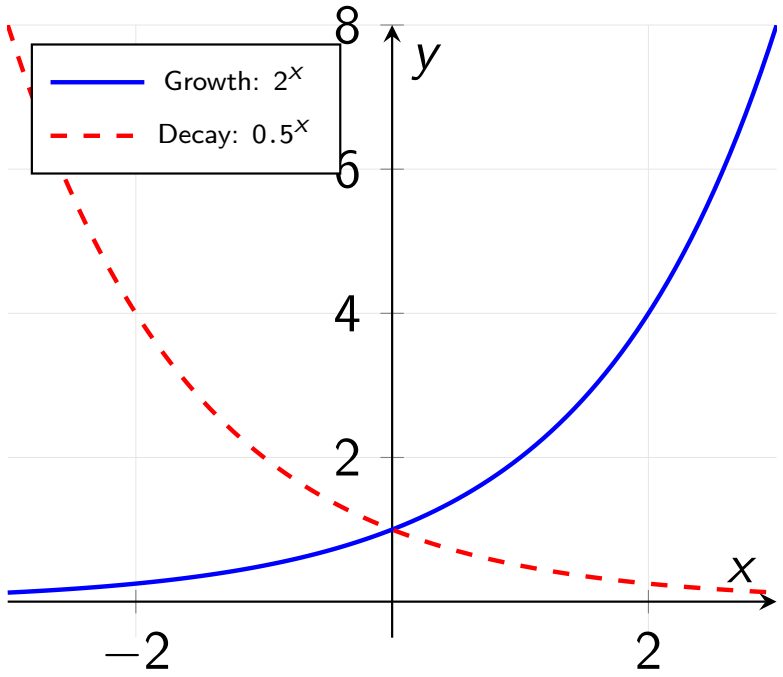
Types of Exponential Functions

Exponential Growth ($a > 1$):

- Function **increases** as x increases.
- Grows rapidly.
- Examples: $f(x) = 2^x$, $f(x) = e^x$.

Exponential Decay ($0 < a < 1$):

- Function **decreases** as x increases.
- Decays toward zero.
- Examples: $f(x) = \left(\frac{1}{2}\right)^x$.



The Natural Exponential Function

Definition

The **natural exponential function** is:

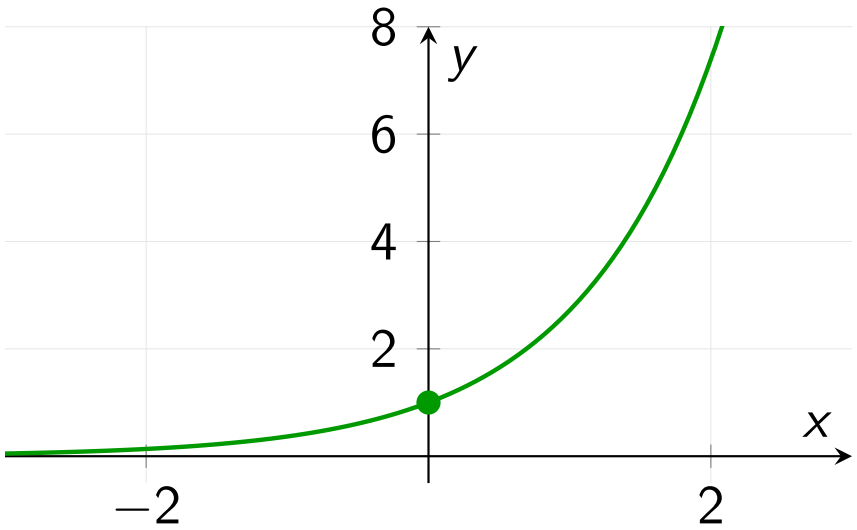
$$f(x) = e^x$$

where $e \approx 2.71828$ is Euler's number, an irrational constant.

Why is e special?

- The slope of the tangent line to $y = e^x$ at $x = 0$ is exactly 1.
- It appears naturally in calculus, compound interest, and population growth.
- $\frac{d}{dx}(e^x) = e^x$ — its own derivative!

The Natural Exponential Function



Properties of Exponential Functions

Laws of Exponents

For $a > 0$, $a \neq 1$, and real numbers m, n:

$$a^m \cdot a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n}$$

How to Graph Exponential Functions

Step-by-Step Method:

1. **Identify the base** a and the parent function $y = a^x$.
2. **Determine growth or decay:**
   - $a > 1$: Growth (increasing).
   - $0 < a < 1$: Decay (decreasing).
3. **Find key points:**
   - y-intercept: $(0, 1)$ for $y = a^x$.
   - Additional points: $(-1, 1/a)$, $(1, a)$, $(2, a^2)$.
4. **Draw the horizontal asymptote:** $y = 0$.
5. **Plot points and connect** with a smooth curve.
6. **Apply transformations** if the function is in the form $y = a^{x-h} + k$.

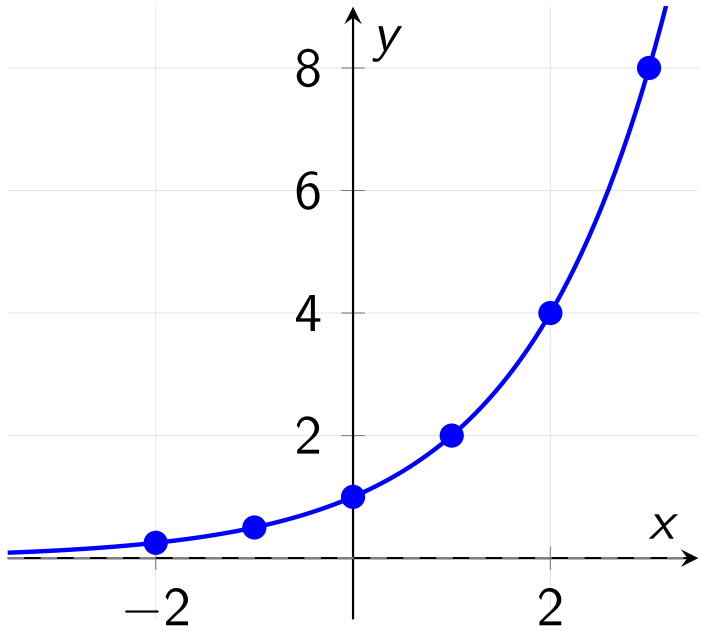
Example 1: Graph of $f(x) = 2^x$

Key Features:

- Base $a = 2 > 1$: Growth.
- y-intercept: $(0, 1)$.
- Asymptote: $y = 0$.
- Domain: $\mathbb{R}$.
- Range: $(0, \infty)$.

Table of Values:

x	$y = 2^x$
-2	0.25
-1	0.5
0	1
1	2
2	4
3	8



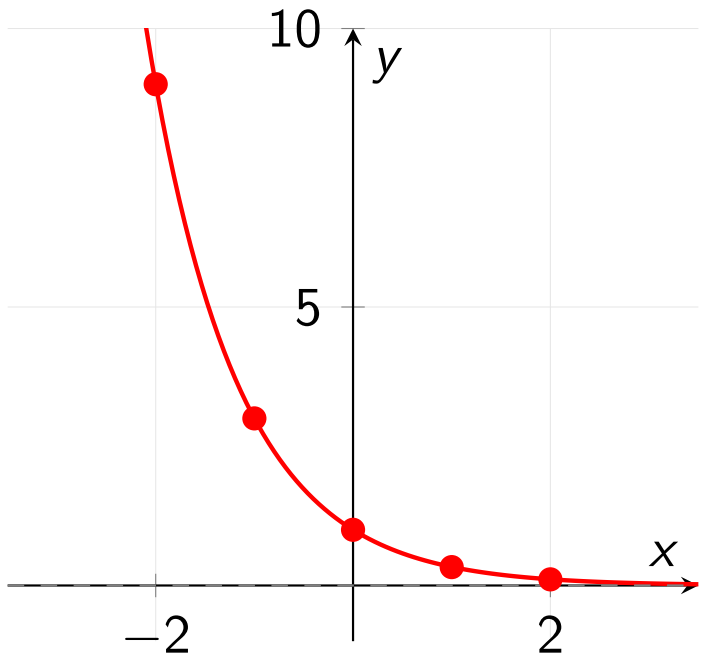
Example 2: Graph of $f(x) = \left(\frac{1}{3}\right)^x$

Key Features:

- Base $a = \frac{1}{3} < 1$: Decay.
- y-intercept: $(0, 1)$.
- Asymptote: $y = 0$.
- Domain: $\mathbb{R}$.
- Range: $(0, \infty)$.

Table of Values:

x	$y = (1/3)^x$
-2	9
-1	3
0	1
1	0.33
2	0.11



Example 3: Graph of $f(x) = e^x$

Key Features:

- Base $a = e \approx 2.718 > 1$: Growth.
- y-intercept: $(0, 1)$.
- Asymptote: $y = 0$.
- Steeper than 2^x.
- Domain: $\mathbb{R}$, Range: $(0, \infty)$.

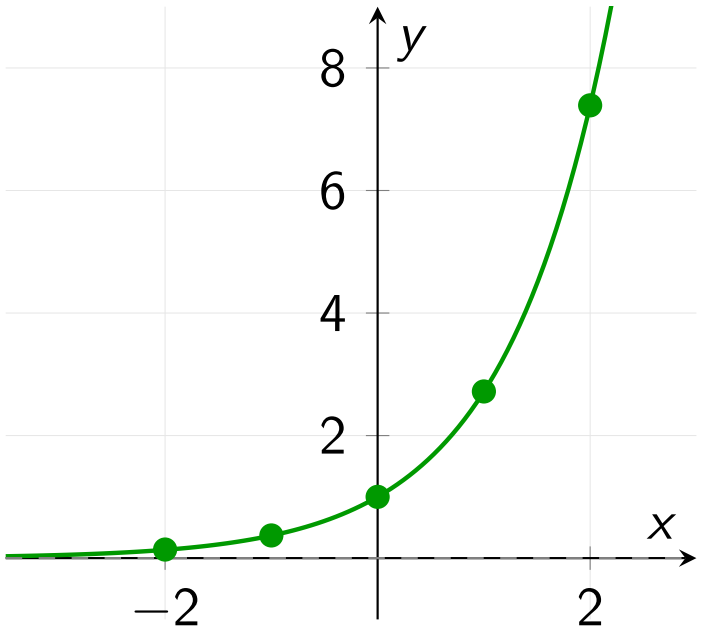


Table of Values:

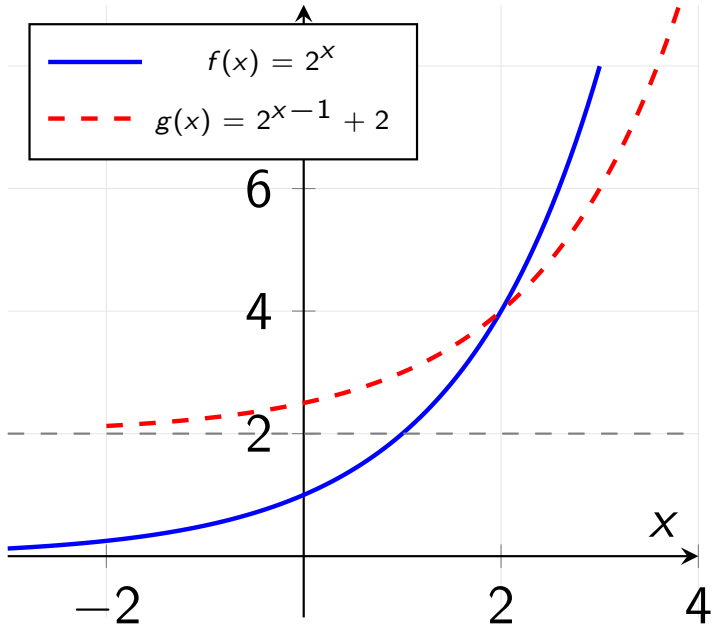
x	$y = e^x$
-2	0.14
-1	0.37
0	1
1	2.72
2	7.39

Example 4: Transformations of Exponential Functions

Graph $f(x) = 2^x$ **and** $g(x) = 2^{x-1} + 2$.

Transformations:

- Start with $f(x) = 2^x$.
- 2^{x-1}: Shift **right** by 1 unit.
- $+2$: Shift **up** by 2 units.
- New asymptote: $y = 2$.
- New y-intercept: $(0, 2.5)$.



Key Points:

x	2^x	$2^{x-1} + 2$
-1	0.5	1.25
0	1	2.5
1	2	3
2	4	4
3	8	6

Example 5: Graph of $f(x) = -2^x + 3$

Transformations:

- Start with $y = 2^x$.
- $-$ sign: Reflect over x-axis.
- $+3$: Shift up by 3 units.
- Asymptote: $y = 3$.
- y-intercept: $(0, 2)$.
- Range: $(-\infty, 3)$.

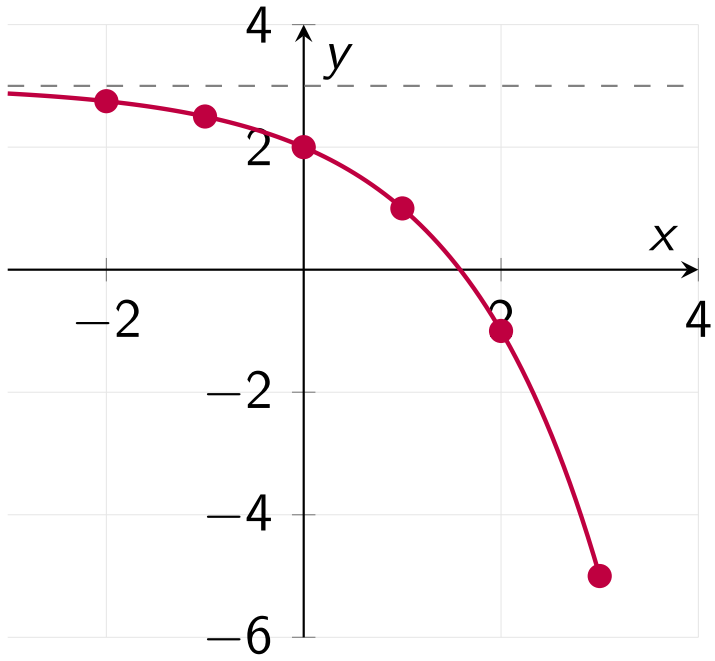


Table of Values:

x	$-2^x + 3$
-2	2.75
-1	2.5
0	2
1	1
2	-1